

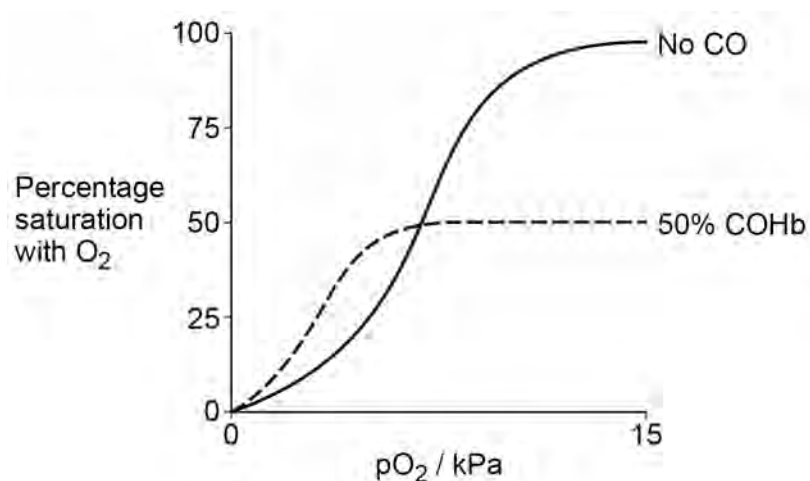
07.1

Carbon monoxide (CO) is released during incomplete combustion of fossil fuels.

Figure 7 shows the dissociation curve for oxyhaemoglobin when:

- **not** exposed to CO
- exposed to CO such that 50% of the oxygen binding sites are occupied by CO (50% COHb).

Figure 7



Using **Figure 7**, what can you conclude about how exposure to CO affects the loading and unloading of oxygen by haemoglobin?

Explain your answer.

[3 marks]

07.2

The World Health Organisation (WHO) suggests that to avoid long-term health effects, COHb concentrations should be kept below 2.5%.

WHO recommends that people should not be exposed to:

- air with > 10 mg m⁻³ CO for more than 8 hours
- air with > 30 mg m⁻³ CO for more than 1 hour.

Scientists have used a mathematical model to calculate the exposure to carbon monoxide that would result in 2.5% COHb in both adults and children.

Table 2 shows the scientists' results.

Table 2

Exposure duration at rest / hours	CO concentration in the air / mg m ⁻³ resulting in 2.5% COHb	
	Child	Adult
1	31.2	40.2
8	9.6	10.6

The scientists suggest that the WHO recommendations for carbon monoxide concentrations resulting in 2.5% COHb should be reduced.

Evaluate the scientists' conclusion.

[3 marks]

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Turn over for the next question

